

Tutorial n: 02 "Optimal Spanning Tree problem & applications"

Exercise 01

Given the following network topologies,

1. How many minimum spanning trees are possible for each case?
2. Which ones are they? Justify your answer.

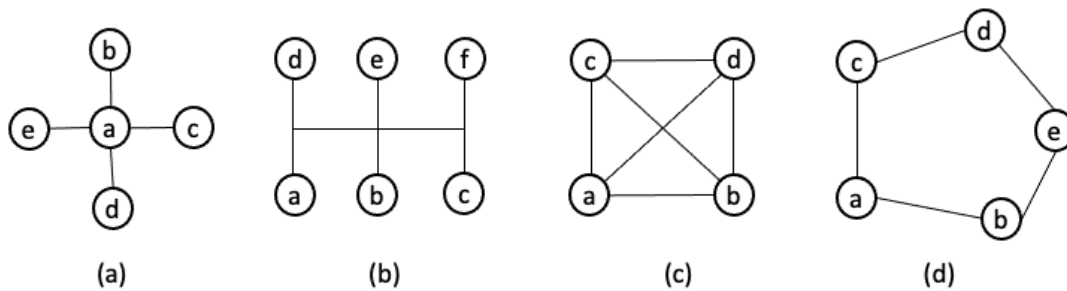


FIGURE 1 –

Exercise 02

Given the following computer network :

1. Construct the minimum spanning tree using Kruskal's algorithm, considering A as its root (Root Bridge).
2. Construct the minimum spanning tree using Prim's algorithm, also considering A as its root (Root Bridge). What difference do you observe?
3. Apply the Spanning Tree Protocol (STP) by unfolding its algorithm on this network. What do you notice?
4. If we replace the routers with sensors and the links with wireless connections, and if we consider the energy consumption status given by (a=90%, b=80%, c=95%, d=60%, e=30%) respectively, what would be the spanning tree of this network?

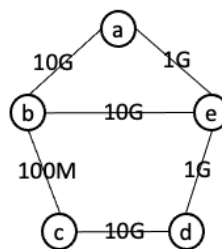


FIGURE 2 – Router network